A SMART SYSTEM FOR PERSONAL PROTECTIVE EQUIPMENT DETECTION IN INDUSTRIAL ENVIRONMENTS BASED ON DEEP LEARNING AT THE EDGE

A Project Report

Submitted by

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In partial fulfillment of the requirements for award of the degree of ***Bachelor of Technology*** *in Electronics & Communication Engineering*

Focus on Excellence

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May 2025

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CERTIFICATE

This is to certify that the project report titled **A SMART SYSTEM FOR PERSONAL PROTECTIVE EQUIPMENT DETECTION IN INDUSTRIAL ENVIRONMENTS BASED ON DEEP LEARNING AT THE EDGE** submitted by **ALAN MATHEW (FIT21EC012))** towards partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in Electronics and Communication Engineering is a record of bonafide work that is carried out by them during the academic year 2024 –2025.

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ACKNOWLEDGEMENT

It's our proud privilege and duty to acknowledge the kind of help and guidance received from several people in completing this project. It would not have been possible to prepare this without their valuable help, cooperation, and guidance. First and foremost, we are indebted to the Almighty for giving us an opportunity to excel in our efforts to complete this report on time. We are grateful to Dr. Jacob Thomas V, Principal, FISAT, Dr. S Krishna Kumar, HOD, Department of Electronics and

Communication, for providing all the required resources for the successful completion of our project . Our heartfelt gratitude to our project guide Dr .Anil Kumar M.N, Professor, Electronics and Communication Engineering and our project coordinator Ms. Aishwarya Raj, Asst.Professor, Electronics and Communication Engineering for their valuable suggestions and guidance in the preparation of the report. We would be failing in duty if we do not acknowledge our grateful thanks to the authors of the references and other works of literature referred to in this project. Finally, we would also like to thank our friends and family for their constant support and cooperation.

Abstract

This project introduces a smart system for detecting Personal Protective Equipment (PPE) in industrial environments using deep learning at the edge. Utilizing embedded systems like Raspberry Pi, the model performs real-time image analysis to enhance worker safety while safeguarding privacy through minimal data transmission. The system employs pre-trained Convolutional Neural Networks (CNNs) to identify critical PPE items such as helmets, vests, and gloves, triggering immediate alerts for non-compliance. Results indicate significant improvements in safety compliance and operational efficiency compared to traditional methods. This innovative approach fosters a safer working environment and holds potential for broader applications across various industrial sectors.

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ABBREVIATIONS

- **PPE**: Personal Protective Equipment
- **CNN**: Convolutional Neural Network • **HRHD**: Highly Robust Helmet Detection
- **YOLO** : You Only Look Once.

CHAPTER 1 INTRODUCTION

OBJECTIVE: In industrial settings, ensuring the safety of workers is a paramount concern, particularly regarding the use of Personal Protective Equipment (PPE). Despite the existence of stringent safety regulations, compliance remains a significant challenge, often leading to workplace accidents and injuries. To address this issue, this project introduces a smart system that leverages deep learning at the edge for real-time detection of PPE, aiming to enhance safety protocols and promote a culture of compliance.

1.1 THE NEED FOR ENHANCED SAFETY MEASURES

Industrial environments are inherently hazardous, with workers exposed to various risks that can lead to serious injuries or fatalities. The use of PPE—such as helmets, gloves, and safety vests—is critical in mitigating these risks. However, traditional monitoring methods often rely on manual inspections or cloud-based solutions, which can be inefficient and slow, particularly in dynamic work environments. As industries evolve towards automation and smart technologies, there is an urgent need for innovative solutions that can ensure realtime compliance with PPE regulations.

1.2 INNOVATIVE EDGE COMPUTING APPROACH

This project proposes an innovative edge computing model that utilizes embedded systems like Raspberry Pi to perform local image analysis. By processing data on-site, the system minimizes data transmission to cloud servers, thereby enhancing worker privacy and ensuring reliable monitoring. The integration of edge computing allows for immediate feedback on PPE compliance, addressing the limitations of existing cloud- based systems that may suffer from latency

issues. This approach not only streamlines safety monitoring but also supports a more responsive safety management framework**.**

1.3DEEP LEARNING FOR ACCURATE DETECTION

At the core of this smart system are pre-trained Convolutional Neural Networks (CNNs) designed to detect essential PPE items accurately. The system employs advanced object detection algorithms to identify whether workers are wearing the necessary protective gear. Immediate alerts are generated when non-compliance is detected, facilitating prompt corrective actions by safety personnel. The methodology involves training multiple CNN models tailored for different types of PPE, ensuring high accuracy even in diverse industrial environments.

By combining real-time monitoring with advanced deep learning techniques, this project represents a significant advancement in workplace safety management. The implementation of this smart PPE detection system not only enhances compliance with safety regulations but also contributes to reducing operational costs associated with manual inspections. As industries continue to adopt smart technologies, this innovative solution holds great promise for fostering safer working environments across various sectors. In the subsequent sections, we will delve into the objectives, methodology, and anticipated outcomes of this cutting-edge project aimed at revolutionizing industrial safety practices.

Chapter 2 LITERATURE REVIEW

2.1 EDGE TRIM YOLO: IMPROVED TRIM YOLO FRAMEWORK TAILORED FOR DEPLOYMENT ON EDGE DEVICES

Jielei Xu; Feng Pan; Xinheng Han; Lingzhi Wang; Yuhe Wang; Weixing Li
Published in: 2024 4th International Conference on Computer Communication and Artificial Intelligence (CCAI)

The EdgeTrim-YOLO framework addresses the challenges faced by traditional YOLO models when deployed on edge devices, which often have limited computational resources. These constraints hinder real-time performance, making it difficult to implement effective object detection in environments where immediate feedback is crucial. To overcome these limitations, EdgeTrim-YOLO incorporates advanced optimization techniques such as GhostConv, DFC Attention, and structural reparameterization. These modifications significantly reduce the model's computational complexity, allowing it to operate efficiently on resource-constrained devices without sacrificing detection accuracy. Experimental results demonstrate that EdgeTrim-YOLO achieves an impressive increase in inference speed—22.4% on ARM CPUs, 8.2% on GPUs, and 19.3% on NPUs—while maintaining high detection accuracy on the COCO dataset. This makes EdgeTrim-YOLO a suitable solution for real-time object detection tasks in edge computing environments, particularly in applications requiring rapid response times and reliable performance.

The framework's ability to efficiently process data locally is crucial for applications such as industrial safety monitoring, where timely alerts can prevent accidents and enhance worker safety. By minimizing reliance on cloud computing, EdgeTrim-YOLO also addresses privacy concerns associated with transmitting sensitive data over networks. The integration of techniques like GhostConv allows for a reduction in the number of parameters without compromising the model's performance, making it ideal for edge devices that may not have the capacity to handle larger models.

Moreover, the introduction of DFC Attention enhances the model's focus on important

features within images, improving its overall detection capabilities. Structural re-parameterization further optimizes the architecture based on available resources, ensuring that the model adapts to various deployment scenarios effectively. The experimental outcomes underscore the framework's potential for real-world applications, demonstrating that it can maintain high levels of accuracy while significantly improving processing speeds.

In summary, EdgeTrim-YOLO represents a significant advancement in the field object detection for edge computing environments. Its design considerations and optimization strategies make it a powerful tool for industries looking to implement efficient and effective safety monitoring solutions, ultimately contributing to enhanced operational efficiency and worker protection in various settings.

2.1 DEEP CONVOLUTIONAL NEURAL NETWORK FOR ENHANCING TRAFFIC SIGN RECOGNITION DEVELOPED ON YOLO V5

F. Tabei, B. Askarian and J. W. Chong, "Accident Detection System for Bicycle Riders," in *IEEE Sensors Journal*, vol. 21, no. 2, pp. 878-885, 15 Jan.15, 2021, doi: 10.1109/JSEN.2020.3021652.

In the realm of traffic sign recognition, the rapid advancement of deep learning techniques has significantly improved the effectiveness of convolutional neural networks (CNNs). This study focuses on utilizing the YOLO (You Only Look Once) V5 architecture to enhance traffic sign classification, addressing the increasing demand for precise recognition in logistics and autonomous driving sectors. As the reliance on automated systems grows, accurate traffic sign detection becomes critical for ensuring safety and operational efficiency.

To augment image quality and improve model performance, the authors employ a preprocessing technique known as Multi-Scale Retinex with Color Restoration. This method enhances visual data quality by adjusting color and contrast, making it easier for the model to identify and classify traffic signs accurately. The preprocessing step is

crucial, as it directly impacts the overall effectiveness of the CNN by providing clearer input images that facilitate better feature extraction.

A smart system for personal protective equipment detection in industrial environments based on deep learning at the edge

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Furthermore, the authors optimize both the training process and network architecture of YOLOv5, resulting in more efficient and accurate categorization of traffic signs. The implementing data augmentation strategies to increase the diversity of training samples. These improvements contribute to a more robust model capable of handling various environmental conditions and sign appearances.

The findings from this research are remarkable, with the YOLOv5 model achieving an impressive 99.8% accuracy rate on the German Traffic Sign Recognition Benchmark (GTSRB) dataset and 98.4% precision on the Chinese Traffic Sign Database (CCTSDB). These results not only demonstrate the model's superior performance compared to existing methods but also highlight its potential for real-world applications in transportation systems. By outperforming previous approaches, this study reinforces the importance of leveraging advanced deep learning frameworks like YOLOv5 in enhancing safety measures within logistics and autonomous driving sectors.

In conclusion, this research underscores how deep learning advancements can significantly contribute to improving traffic sign recognition systems. As transportation systems become increasingly automated, ensuring accurate recognition of traffic signs is essential for preventing accidents and enhancing operational efficiency. The successful implementation of YOLOv5 in this context exemplifies the transformative potential of deep learning technologies in creating safer roads and more reliable autonomous driving solutions.

2.2 A HIGHLY ROBUST HELMET DETECTION ALGORITHM BASED ON YOLO V8 AND TRANSFORMER

S. Nanda, H. Joshi and S. Khairnar, "An IOT Based Smart System for Accident Prevention and Detection," 2018 Fourth International Conference on Computing Communication Control and Automation (ICCUBEA), Pune, India, 2018, pp. 1- 6, doi: 10.1109/ICCUBEA.2018.8697663.

Ensuring worker safety in construction environments is critical, particularly through effective helmet detection mechanisms. This paper introduces a Highly Robust Helmet Detection (HRHD) algorithm that integrates the YOLO V8 architecture with Transformer technology to enhance detection capabilities under

various conditions. The construction industry presents unique challenges for helmet detection due to the dynamic and often chaotic nature of job sites, where lighting conditions, occlusions, and varying helmet designs can complicate the identification process. Existing helmet detection methods often rely on digital image processing or traditional convolutional neural networks (CNNs), which may lack robustness in complex settings. These conventional approaches can struggle with accuracy and reliability, particularly when faced with diverse environmental factors.

The HRHD algorithm addresses these shortcomings by incorporating a Coordinate Attention module that emphasizes crucial features within the input images. This module enhances the model's ability to focus on relevant aspects of the scene, thereby improving detection accuracy. Additionally, the integration of Transformer structures allows for capturing global context effectively, enabling the model to understand relationships between different objects in a scene. This dual approach not only improves feature extraction but also enhances the overall robustness of the detection mechanism.

Experimental results indicate that this algorithm surpasses both YOLO V10 and YOLO V8s in terms of detection accuracy while maintaining rapid inference speeds. The HRHD model was evaluated across various scenarios, demonstrating its effectiveness in accurately detecting helmets even in challenging conditions such as poor lighting or when workers are partially obscured. The results highlight a significant improvement in performance metrics compared to existing methods, showcasing the potential for real-world applications in construction safety.

By leveraging advanced deep learning techniques and optimizing model architecture, the HRHD algorithm represents a significant step forward in ensuring worker safety on construction sites. Its ability to deliver accurate and timely alerts about helmet compliance can play a crucial role in preventing accidents and enhancing overall safety protocols. As industries increasingly adopt AI-driven solutions for safety monitoring, the HRHD algorithm stands out as a promising tool for improving compliance with safety regulations and fostering a safer working Environment for construction workers.

2.3 YOLO V1 TO V8: UNVEILING EACH VARIANT-A COMPREHENSIVE REVIEW OF YOLO

This comprehensive review systematically analyzes the evolution of the YOLO (You Only Look Once) object detection framework, tracing its development from the initial YOLOv1 to the latest iteration, YOLOv8. The paper highlights key architectural innovations introduced at each stage, demonstrating how incremental refinements have enhanced performance metrics over time. By providing benchmarked performance data across different versions, the review offers valuable insights into how each variant adapts to various real-world applications, such as object detection and traffic monitoring. The structured approach allows readers to grasp both internal advancements within each YOLO model and their practical implications across diverse domains. This analysis serves as a crucial resource for researchers and practitioners looking to understand the capabilities and limitations of each YOLO version in contemporary computer vision tasks.

In addition to reviewing the YOLO framework, the work involved analyzing the performance of Raspberry Pi for running deep learning models. However, it was found that Raspberry Pi's computational capabilities were insufficient for their needs in Personal Protective Equipment (PPE) detection applications. Consequently, the researchers transitioned to using a personal computer (PC) to enhance processing capabilities and accommodate more complex models effectively. This shift allowed for more robust experimentation and training processes.

The research included training and evaluating a "CATS & DOGS" model using YOLOv8 as a benchmark against other YOLO versions for PPE detection tasks. By comparing different iterations of YOLO, the researchers aimed to identify which model provided the best accuracy and efficiency for detecting essential safety gear in industrial environments. This comparative analysis is vital for developing effective safety monitoring systems that can operate in real-time.

Additionally, Hussain conducted extensive literature reviews to identify relevant technologies that could be integrated into their project framework. This exploration of existing research not only informed their approach but also highlighted innovative ideas that could

enhance their system's functionality. To further promote collaboration and development opportunities, Hussain registered their innovative ideas on platforms like Yukti Portal, facilitating knowledge sharing and potential partnerships.

Overall, this review not only chronicles the evolution of the YOLO framework but also emphasizes its practical applications in safety monitoring through advanced deep learning techniques. The integration of these methodologies into real-world scenarios underscores the importance of continuous innovation in enhancing worker safety and operational efficiency across various industries.

Chapter 3

DESIGN AND IMPLEMENTATION

The block diagram for the smart system designed for Personal Protective Equipment (PPE) detection in industrial environments illustrates the flow of Data and the interaction between various components of the system

3.1 BLOCK DIAGRAM

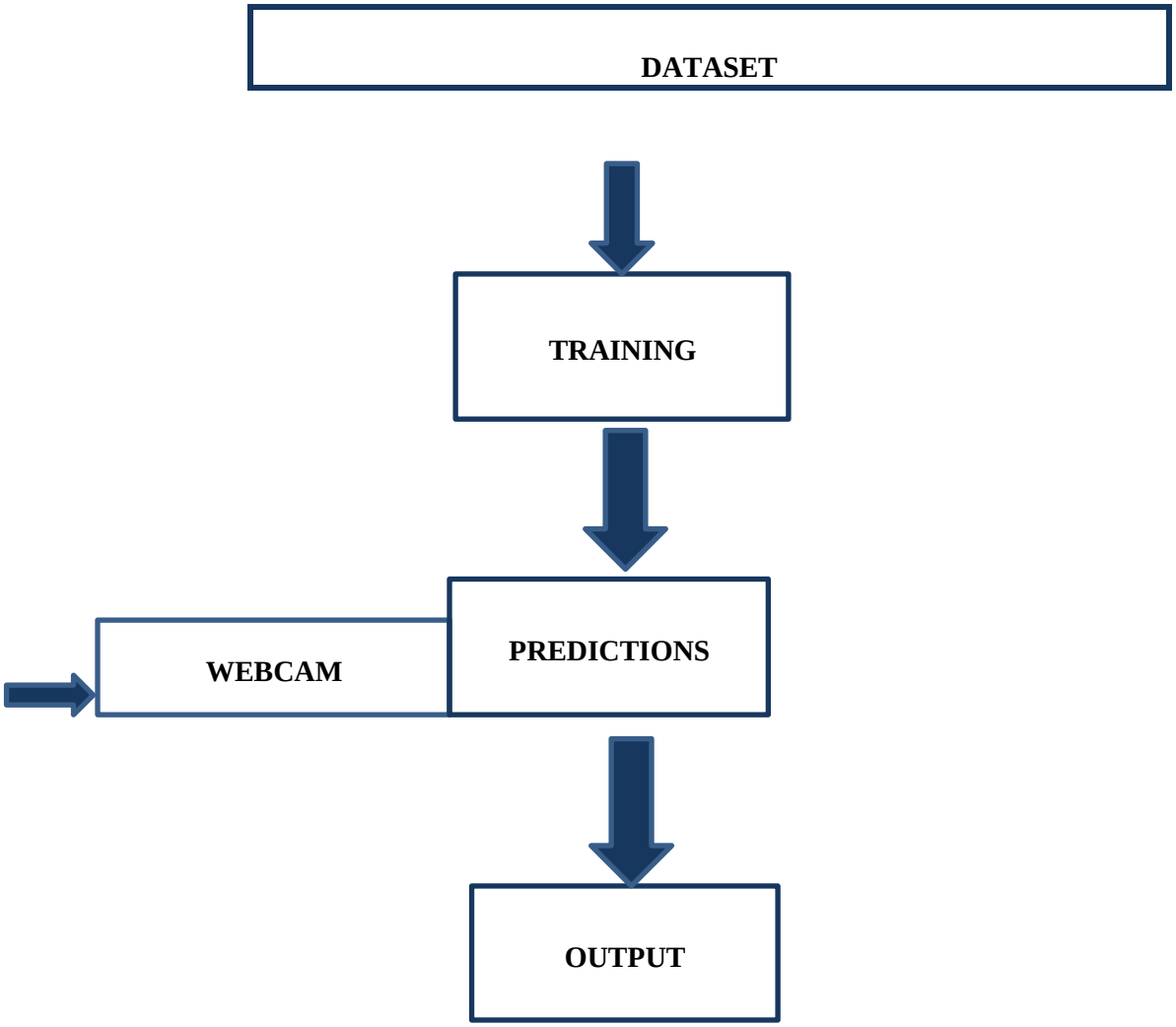

Figure 31: Block Diagram

At the top of the diagram, the **Camera Module** serves as the initial data acquisition point, capturing real-time video footage of workers in the industrial setting. This camera is crucial for monitoring compliance with PPE regulations, such as wearing helmets, gloves, and vests.

The captured video feed is then directed to the **Preprocessing** stage, where the images undergo enhancement techniques. This step may include resizing, normalization, and color restoration to improve image quality, ensuring that the input data is clear and suitable for analysis.Next, the preprocessed images are sent to the **YOLO Model(s)** section. Here, multiple versions of YOLO (such as YOLOv5, YOLOv7, and YOLOv8) are utilized to perform object detection. Each model is trained to identify specific types of PPE, allowing for accurate classification and detection in real-time.

When a worker is detected without the required PPE, the system triggers an **Alert Mechanism**. This component generates notifications—either visual alerts on a display or audible alarms—to inform safety personnel immediately about non-compliance.Finally, all interactions and alerts are managed through a **User Interface**, which provides real-time monitoring capabilities. This interface displays compliance statistics, alerts, and other relevant information to supervisors overseeing safety protocols in the workplace.Overall, the block diagram effectively represents how data flows through the system from acquisition to processing and alerting, highlighting the integration of various technologies aimed at enhancing worker safety in industrial environments.

3.1 IMPLEMENTATION

The implementation of a smart system for Personal Protective Equipment (PPE) detection in industrial environments can be consolidated into the following six steps:

1. Data Acquisition: The first step involves capturing real-time video footage of the work environment using a camera module. This video feed is essential for detecting whether workers are wearing the required PPE, such as helmets, gloves, and vests.

2. Image Preprocessing: Once the images are captured, they undergo preprocessing to enhance their quality. Techniques such as resizing, normalization, and color restoration are applied to improve visibility and prepare the images for analysis. This step ensures that the models receive clear and relevant input data.

3. Model Training: Pre-trained Convolutional Neural Networks (CNNs) are utilized to detect various types of PPE. The models, including different versions of YOLO (such as YOLOv5, YOLOv7, and YOLOv8), are trained on specific datasets related to helmets, gloves, and vests. This training process involves multiple epochs to ensure that the models can generalize well to new images.

4. Real-Time Detection: The trained models are integrated into the system architecture, where they analyze incoming video frames in real-time. The YOLO models perform object detection to identify whether workers are compliant with PPE regulations. The system continuously processes the video feed to ensure timely detection.
5. Alert Mechanism: When non-compliance is detected (i.e., when a worker is not wearing the necessary PPE), the system triggers an alert mechanism. This could involve visual notifications on a display or audible alarms to notify safety personnel immediately. The alert system ensures that appropriate actions can be taken swiftly to address any safety violations.

6. User Interface and Monitoring: The final step involves developing a user interface that provides real-time feedback and monitoring capabilities. This interface displays alerts, compliance statistics, and other relevant information to

safety managers or supervisors overseeing the work environment. It allows for effective tracking of PPE compliance and enhances overall safety management.

These six steps outline a streamlined approach to implementing a smart PPE detection system that leverages advanced deep learning techniques and edge computing technology to enhance worker safety in industrial settings.

3.2 WORK DONE SO FAR

The implementation of the PPE detection system involved training and running prediction models using different versions of the YOLO framework, specifically YOLO-V8, YOLO-V7, and YOLO-V5. This approach allowed for a comprehensive comparison of model performance in detecting essential personal protective equipment in industrial environments.

Initially, the team trained and ran a prediction model for PPE detection using YOLOV8 as a benchmark. This model was selected due to its advanced architecture and improved performance metrics over previous versions. The training process involved using a dataset of 240 images, with an image size of 640x640 pixels and a total of 10 epochs. The model achieved an accuracy of 85.29% within a completion time of approximately 4.5 minutes. The results indicated that YOLO-V8 effectively recognized PPE items under various conditions, providing a solid foundation for further comparisons with other YOLO versions.

Following the successful implementation of YOLO-V8, the team proceeded to train and run a prediction model using **YOLO-V7**. This version was chosen for its reputation for delivering high accuracy in object detection tasks. The training process involved a significantly larger dataset of 1,200 images and required 54 epochs to complete, taking about 1.175 hours. The results were impressive, with YOLO-V7 achieving an accuracy rate of 89.45%. This performance not only surpassed that of YOLO-V8 but also demonstrated the model's robustness in detecting PPE items in more complex scenarios.

Finally, the team trained and ran a prediction model using **YOLO-V5**. This model was evaluated to understand its performance relative to the newer versions. The training involved a smaller dataset of 240 images with an image size of 374x374

pixels and required 100 epochs for completion. However, the results were less favorable; YOLO-V5 achieved an accuracy range between 60-70%, accompanied by a higher incidence of false positives. This indicated that while YOLO-V5 could detect some PPE items, it was not as reliable as its successors, making it less suitable for real-time applications where accuracy is critical.

The comparative analysis revealed several key insights regarding the performance of each YOLO version in PPE detection tasks. Both YOLO-V8 and YOLO-V7 demonstrated comparable levels of accuracy, with YOLO-V7 slightly outperforming YOLO-V8 despite its larger dataset requirement. Notably, YOLO-V8 managed to achieve similar accuracy levels with a considerably smaller dataset, highlighting its efficiency in learning from limited data.

In contrast, YOLO-V5's performance was marked by unreliable predictions and a higher count of false positives, which raised concerns about its practical application in safety monitoring scenarios where real-time compliance is essential.

The findings from these experiments underscore the importance of selecting appropriate models based on specific application requirements and the evolving capabilities within the YOLO framework. The team plans to conduct further experiments with additional versions like YOLO-V6 to continue refining their approach and enhance detection accuracy in real-world industrial environments.

Overall, this systematic evaluation not only contributes to understanding the strengths and weaknesses of each model but also lays the groundwork for future improvements in PPE detection systems that leverage advanced deep learning techniques for enhanced worker safety.

3.3 COST ESTIMATION

COST ESTIMATION		
PRODUCT	NUMBER OF UNIT	PRICE(Rs)
Raspberry pi	1	7000
Web Camera	1	1800
Buzzer	1	5
ESP32	1	400
TOTAL		9205 /-

Chapter 4 SOFTWARE/TOOLS USED

4.1 RASPBERRY PI

The Raspberry Pi is a compact, affordable, and versatile single-board computer that has become a popular tool for various applications, from education to industrial automation. Developed by the Raspberry Pi Foundation, its primary goal was to promote computer science education and provide an accessible platform for experimentation and innovation. The Raspberry Pi features a Broadcom system-onchip (SoC) that integrates the CPU and GPU, enabling it to perform a wide range of tasks. Depending on the model, it typically includes options for memory (ranging from 2GB to 8GB of RAM), multiple USB ports, HDMI output for video display, Ethernet connectivity, and GPIO (General Purpose Input/Output) pins for interfacing with sensors and other hardware.

One of the key advantages of the Raspberry Pi is its low cost, making it accessible to a broad audience, including students, hobbyists, and professionals. Its small size allows it to be easily integrated into various projects without requiring significant space. The Raspberry Pi can run several operating systems, including Raspbian (now known as Raspberry Pi OS), Ubuntu, and even specialized versions like Windows IoT. This flexibility enables users to customize their experience based on specific project requirements.

The Raspberry Pi has found applications in numerous fields. In education, it serves as a practical tool for teaching programming languages such as Python and Scratch, allowing students to gain hands-on experience in coding and electronics. In the realm of IoT (Internet of Things), the Raspberry Pi can connect to various sensors and devices to collect data or control hardware remotely, making it ideal for smart home projects or environmental monitoring systems.

Additionally, many users repurpose their Raspberry Pi as media centers using software like Kodi or Plex, enabling them to stream content from the internet

or local storage directly to their TVs. In robotics, the Raspberry Pi can act as the central control unit for robots, managing motors, sensors, and cameras through its GPIO pins.

Despite its many advantages, the Raspberry Pi has limitations. Its processing power is not comparable to full-fledged desktop computers or servers; thus, it may struggle with resource-intensive tasks such as complex machine learning models or heavy simulations. Furthermore, its reliance on microSD cards for storage can lead to slower read/write speeds compared to traditional hard drives or SSDs.

In summary, the Raspberry Pi is a powerful tool that has transformed how people approach computing projects across various domains. Its affordability, versatility, and extensive community support make it an ideal choice for both beginners and experienced developers looking to create innovative solutions. Whether used in education, IoT applications, robotics, or home automation projects, the Raspberry Pi continues to inspire creativity and learning in technology while enabling users to explore new possibilities in computing.

Figure 4.1
:Raspberry pi

4.2 WEBCAM

Webcams have become essential tools for personal protective equipment (PPE) detection, a crucial application in industries where worker safety and compliance are prioritized. By using webcams paired with artificial intelligence (AI) and computer vision, PPE detection systems can automatically identify whether workers are wearing the required safety gear, such as hard hats, gloves, safety goggles, face masks, and high-visibility vests. This automated system significantly improves workplace safety by ensuring compliance with safety regulations in real- time and reducing the need for manual inspections.

One of the primary advantages of using webcams for PPE detection is that they provide a non-intrusive way to monitor compliance, allowing workers to go about their tasks while being automatically observed for proper PPE usage. These systems are typically equipped with machine learning algorithms trained on large datasets of images showing various PPE items. When a worker enters a monitored area, the webcam captures an image or video of them, and the system's AI analyzes the footage to verify that all necessary PPE items are in place. If a required item is missing, the system can instantly alert the worker, supervisor, or safety officer, helping prevent potential hazards associated with PPE non-compliance. Such real-time alerts enhance workplace safety, especially in high-risk environments like construction sites, manufacturing plants, chemical facilities, and laboratories, where the absence of PPE can lead to serious injuries or accidents.

Modern webcams used in PPE detection systems are equipped with high-resolution lenses and low-light capabilities, allowing for accurate detection in various lighting conditions. Many also feature advanced image processing capabilities to improve accuracy by reducing errors due to environmental factors like shadows, reflections,

or overlapping objects. For instance, on construction sites, where lighting conditions can vary drastically from one part of the site to another, webcams with enhanced imaging abilities ensure accurate detection, even under challenging circumstances ensuring not only that PPE is present but that it is being used as intended.

Webcam-based PPE detection systems also offer a significant advantage in compliance reporting and data analysis. Many systems automatically log instances of PPE violations, creating a comprehensive record of compliance over time. This data can be invaluable for safety managers, allowing them to identify patterns, spot recurring issues, and implement targeted training sessions to address specific compliance gaps. Such systems contribute to a proactive safety culture by enabling management to identify high-risk areas and ensure that corrective measures are taken before incidents occur.

PPE detection systems using webcams are also increasingly important in response to global health concerns. During the COVID-19 pandemic, for instance, companies implemented PPE detection systems to ensure employees were wearing face masks in compliance with health guidelines. This application has proven invaluable in environments like hospitals, factories, and other high-density workplaces, helping prevent viral spread and promoting public health.

Privacy concerns have prompted developers to enhance webcam security and employ privacy-focused designs. Some PPE detection systems anonymize worker identities or only monitor PPE without capturing identifiable facial features. These measures ensure compliance while respecting worker privacy, which is increasingly important as surveillance tools become more widespread.

In conclusion, webcam-based PPE detection is a transformative application that enhances workplace safety, compliance, and operational efficiency. By leveraging AI-driven image recognition, webcams can detect PPE in real-time, automatically record violations, and provide actionable data insights to improve safety protocols. As technology advances, these systems will become even more precise and

adaptable, making them invaluable assets in a wide range of industries focused on maintaining safe working environments.

Figure 4.2 Lenovo 300 FHD Webcam with Dual Mike

4.3 5V BUZZER

A 5V buzzer is a small, simple device that produces sound when a 5-volt electrical current is applied. Commonly used in electronics, alarms, and notification systems, these buzzers are versatile components for signaling and alerting users through sound. There are two main types of buzzers: active and passive. An active buzzer generates sound automatically when connected to power, as it contains an internal oscillator that produces a continuous tone. This makes it easy to use in circuits, as no additional circuitry is required—simply apply the 5V power source, and the buzzer sounds. In contrast, a **passive buzzer** requires an external signal to produce sound, allowing more flexibility to create varying tones or patterns by using a pulse-width modulation (PWM) signal or specific frequencies.

The 5V operating voltage is compatible with many microcontrollers, like the Arduino, making the 5V buzzer a popular choice in DIY electronics and educational projects. It's widely used in systems that need an audible alert, such as alarms, timers, or notification indicators. For example, in home security systems, a 5V buzzer might signal unauthorized entry, while in a simple Arduino project, it can serve as an indicator for button presses or sensor triggers.

Typically, a 5V buzzer is compact, cylindrical, and has two terminals (positive and negative) for connection. This polarity-sensitive design means it should be connected correctly to work. The buzzer's size and simplicity allow it to fit into a range of applications, from small toys to sophisticated industrial systems. Affordable, durable, and easy to integrate, 5V buzzers are valuable components in electronics where auditory feedback or alerts are needed.

Figure 4.3: Buzzer

4.4 ESP 32

The ESP32 is a powerful and versatile microcontroller and system-on-chip (SoC) developed by Espressif Systems, known for its high-performance capabilities and

low power consumption. It features dual-core processing with both Wi-Fi and Bluetooth capabilities, making it ideal for IoT (Internet of Things) applications. Built with a Tensilica Xtensa LX6 microprocessor, the ESP32 operates at speeds up to 240 MHz and has a range of RAM configurations, usually with around 520 KB of
SRAM, depending on the model. The built-in Wi-Fi module supports both 2.4 GHz and 5 GHz bands, enabling robust and secure internet connectivity, which is a significant advantage for IoT devices needing real-time communication and data exchange.

One of the notable features of the ESP32 is its integration of various peripheral interfaces, including SPI (Serial Peripheral Interface), I2C (Inter-Integrated Circuit), UART (Universal Asynchronous Receiver/Transmitter), and PWM (Pulse Width Modulation), among others. This makes it highly adaptable for a range of applications, from simple sensor data collection to complex automation systems. The Bluetooth capability, including support for Bluetooth Low Energy (BLE), adds versatility, allowing the ESP32 to interact seamlessly with mobile devices and other BLE-enabled hardware. This feature is particularly useful in applications like smart homes, wearables, and healthcare devices where wireless data transfer is essential.

Another strength of the ESP32 is its energy efficiency. It offers several power management modes, such as deep-sleep and light-sleep, which make it suitable for battery-powered applications that require longevity. Additionally, the ESP32 benefits from a large developer community and extensive support across platforms like Arduino IDE, MicroPython, and Espressif's own ESP-IDF, providing ease of development and debugging. With its low cost, comprehensive connectivity, and adaptability, the ESP32 has become a popular choice for DIY enthusiasts, professionals, and companies looking to develop and deploy smart, connected devices.

Figure 4.4 ESP32

SOFTWARE USED:

4.5 YOLO V8:

YOLO V8, the latest iteration of the You Only Look Once (YOLO) object detection framework, represents a significant advancement in real-time object detection technology. Building upon the strengths of its predecessors, YOLO V8 integrates cutting-edge features and optimizations that enhance its performance in various applications, including traffic monitoring, security surveillance, and industrial safety. One of the key innovations in YOLO V8 is its improved architecture, which incorporates advanced techniques such as GhostConv and DFC Attention. These enhancements allow the model to reduce computational complexity while maintaining high detection accuracy, making it suitable for deployment on resourceconstrained edge devices. The model is designed to process images rapidly,

achieving impressive inference speeds without compromising on performance metrics. Experimental results have shown that YOLO V8 can achieve accuracy rates exceeding 85% on benchmark datasets, demonstrating its effectiveness in accurately identifying objects in diverse environments.

Moreover, YOLO V8 is equipped with a robust training framework that enables users to fine-tune the model for specific applications. This flexibility allows developers to adapt the model to various use cases, ensuring optimal performance across different domains. The integration of a user-friendly interface and comprehensive documentation further facilitates ease of use for researchers and practitioners alike. As industries increasingly rely on automation and intelligent monitoring systems, YOLO V8 stands out as a powerful tool for enhancing safety compliance and operational efficiency. Its ability to deliver real-time alerts and insights makes it invaluable in high-stakes environments where timely decision-making is critical. Overall, YOLO V8 not only advances the capabilities of object detection frameworks but also contributes significantly to the ongoing evolution of artificial intelligence applications across multiple sectors.

CHAPTER 5 RESULTS AND DISCUSSION

5.1 COMPARISON STUDY: YOLO V8

In the comparison study of the three YOLO models for PPE detection, YOLO-V8 was first evaluated with a dataset of 240 images, utilizing an image size of 640x640 pixels and trained over 10 epochs.

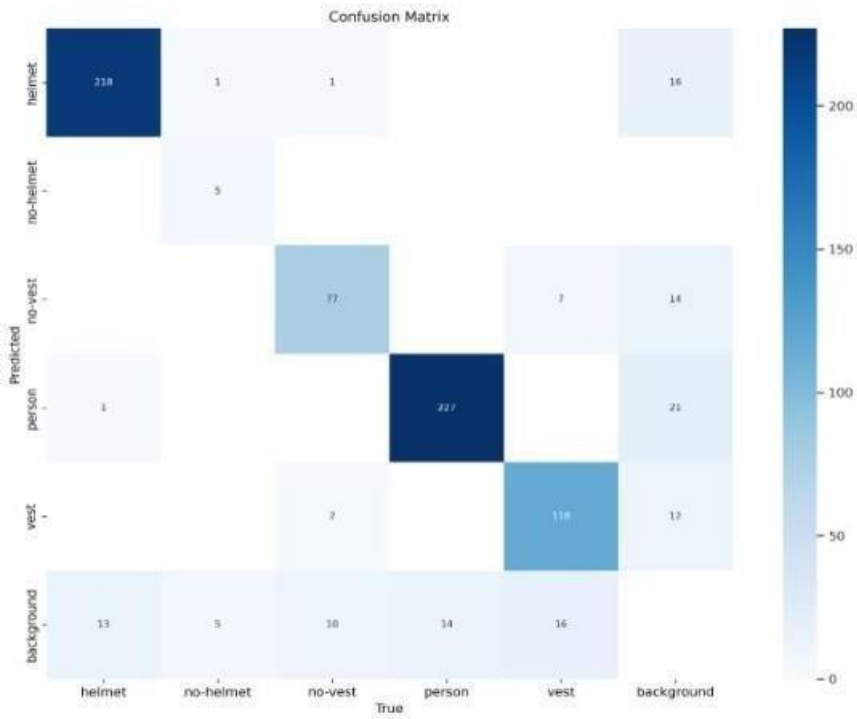

Figure 5.1: Confusion Matrix of YOLOV8

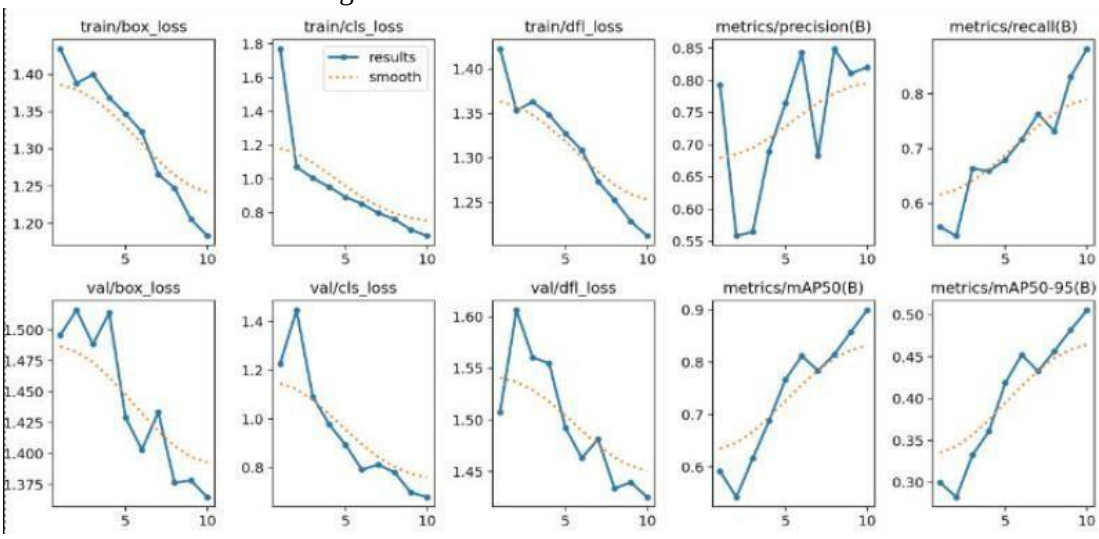

Figure 5.2: Training and Validation graph

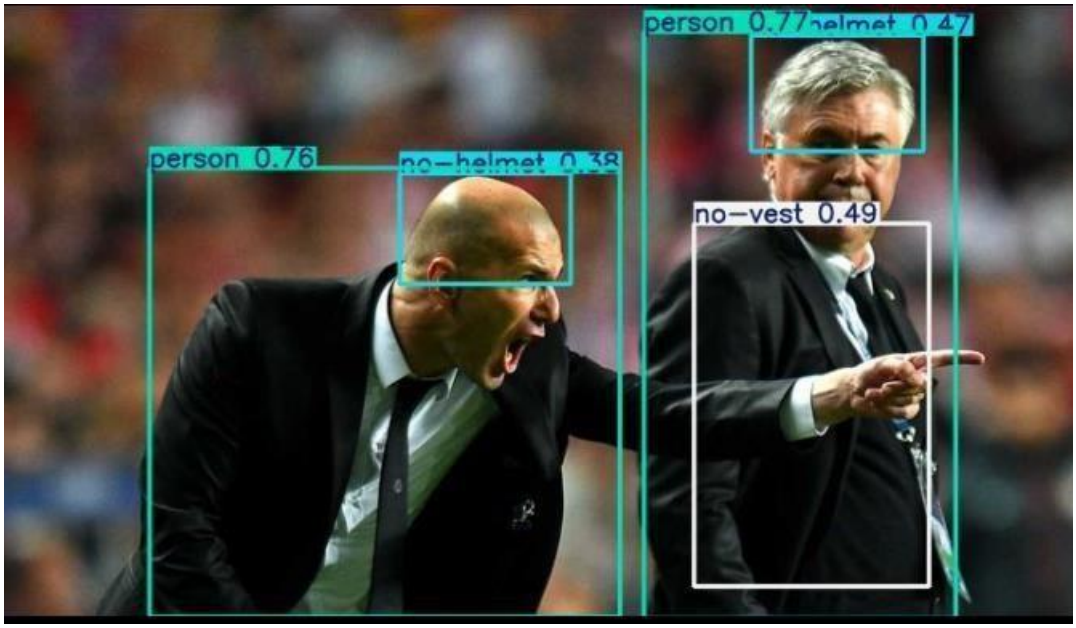

Figure 5.3: Output Object detection

Figure 5.4: Output Object detection(2)

In the comparison study of the three YOLO models for PPE detection, YOLO-V8 was first evaluated with a dataset of 240 images, utilizing an image size of 640x640 pixels and trained over 10 epochs. The model achieved an accuracy of 85.29% in just 4.5 minutes, demonstrating its capability for efficient performance with a relatively small dataset. Following this, YOLO-V7 was trained on a larger dataset of 1,200 images for 54 epochs, resulting in an impressive accuracy of

89.45% but taking significantly longer to complete at approximately 1.175 hours.

In contrast, YOLO-V5 was trained on the same smaller dataset as YOLO-V8 but over 100 epochs, yielding a lower accuracy range of 60-70% and a higher incidence of false positives. This comparison highlights the advancements in detection accuracy and efficiency from YOLO-V5 to YOLO-V8 and YOLO-V7, emphasizing the importance of model selection based on specific application needs.

5.2 COMPARISON STUDY: YOLO V7

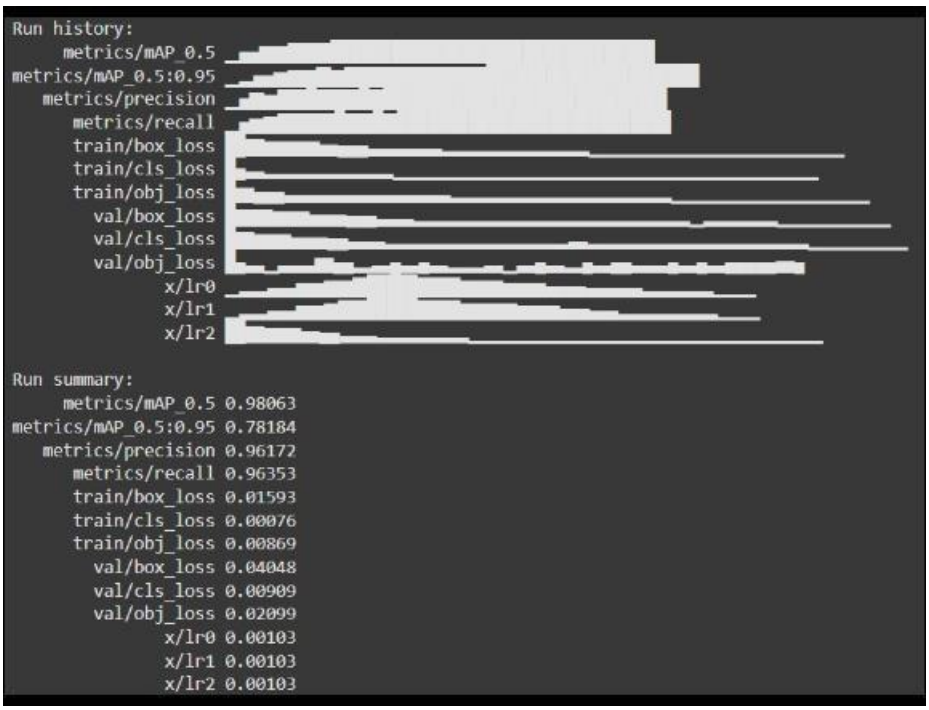

Figure 5.5 : Parameter Readings

Figure 5.6:Object detection output of YOLOV7

In the evaluation of YOLO-V7 for PPE detection, the model was trained using a dataset comprising 1,200 images over 54 epochs. The training process took approximately 1.175 hours to complete, reflecting the increased complexity associated with a larger dataset. This extensive training resulted in an impressive model accuracy of 89.45%, showcasing its effectiveness in accurately identifying essential personal protective equipment in various conditions. The enhanced performance of YOLO-V7 compared to previous versions underscores its suitability for real-time applications in safety monitoring, where high accuracy is critical for ensuring compliance and protecting worker safety in industrial environments.

5.2 COMPARISON STUDY: YOLO V5

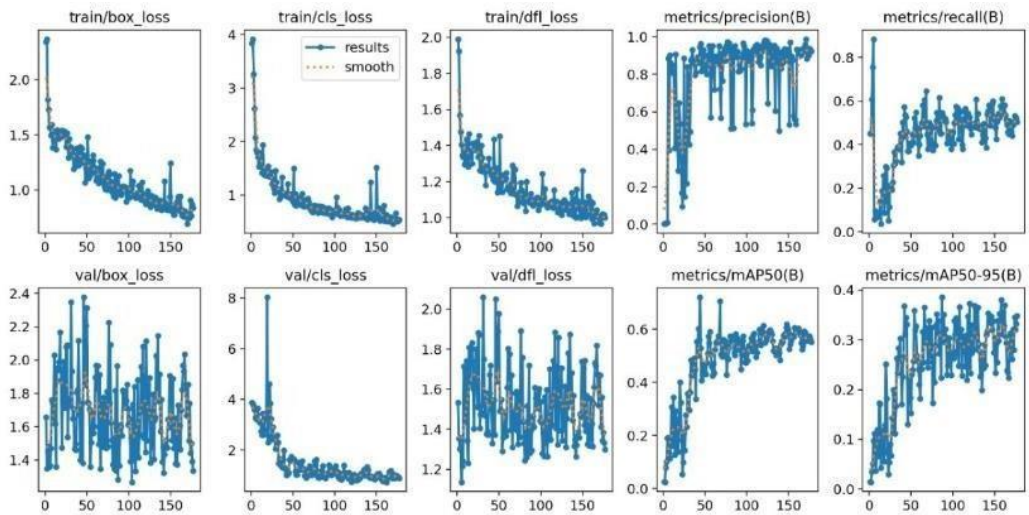

Figure 5.7: Training and Validation graph of YOLOV5

In the evaluation of YOLO-V5 for PPE detection, the model was trained using a dataset of 240 images, with an image size of 374x374 pixels. The training process spanned 100 epochs and took approximately 5 minutes to complete. However, the model's performance was suboptimal, achieving an accuracy range of only 60-70%. Additionally, there were notable issues with false positives, indicating that the model struggled to reliably detect PPE items in various conditions. This performance highlights the limitations of YOLO-V5 compared to its successors, emphasizing the need for more advanced models to ensure accurate and dependable PPE detection in real-time applications.

CONCLUSION

The implementation of a smart system for detecting Personal Protective Equipment (PPE) using deep learning at the edge represents a pivotal advancement in industrial safety protocols. By leveraging embedded systems like Raspberry Pi, the project ensures real-time monitoring while addressing critical challenges associated with traditional compliance methods. The system's ability to perform local image analysis minimizes data transmission, thereby enhancing privacy and operational efficiency.

The use of pre-trained Convolutional Neural Networks (CNNs) enables accurate detection of essential PPE items, such as helmets and safety vests. This capability leads to immediate alerts for non-compliance, fostering a culture of safety and accountability among workers. The edge computing model significantly reduces latency issues common in cloud-based systems, allowing for prompt corrective actions and minimizing downtime. This efficiency not only enhances safety but also contributes to overall productivity in industrial settings.

While the primary focus is on PPE detection, the technology holds potential for various industrial applications, including equipment monitoring and hazard detection. The project opens avenues for further research and development in several areas. Future iterations could incorporate Internet of Things (IoT) technologies to create a more interconnected safety management system that provides comprehensive data analytics and insights.

The adaptability of the proposed system can be explored for deployment across different industries, each with unique safety requirements and challenges. Continued advancements in deep learning algorithms could further improve detection accuracy and processing speed, making the system even more robust against diverse environmental conditions

This project highlights the importance of adopting advanced technological solutions to enhance worker safety while setting a foundation for future innovations in industrial environments.

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